

Topic 2.7 - Composition of Functions

Practice Worksheet

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____

Class: _____

Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. For contextual models, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Evaluate compositions from formulas, tables, graphs, and contexts.
- Determine composition domains and explain why order matters.
- Construct and decompose composite functions while interpreting units and intermediate quantities.

Section A: Foundation and representation

Question 1

Core skill

No calculator

Let $f(x) = 3x - 2$ and $g(x) = x^2 + 4$. Find $(f \circ g)(x)$ and $(g \circ f)(x)$.

Question 2

Core skill

No calculator

Let $p(x) = \frac{1}{x-5}$ and $q(x) = 2x + 1$. Find $(p \circ q)(x)$ and its domain.

Question 3

Core skill

No calculator

Let $f(u) = \sqrt{u+6}$ and $g(x) = x^2 - 10$. Find $(f \circ g)(x)$ and its domain.

Question 4

Core skill

No calculator

The tables define f and g .

x	-2	-1	0	1	2
$f(x)$	2	0	1	-1	-2
$g(x)$	1	2	-1	0	2

Find $(f \circ g)(-2)$, $(g \circ f)(-2)$, and $(f \circ g)(1)$.

Question 5

Decomposition

No calculator

Decompose $H(x) = (5x - 1)^4 + 7$ as $f(g(x))$ in two different ways.

Question 6

Core skill

No calculator

Find all x for which $(f \circ g)(x) = 19$ if $f(u) = 2u + 3$ and $g(x) = x^2 - 1$.

Section B: Application, comparison, and error analysis**Question 7**

Context

No calculator

A printer scales a drawing by factor 1.4 and then adds a 2 cm border to each dimension. Let $S(x) = 1.4x$ and $B(x) = x + 4$ for the total dimension after borders on both sides. Write the composite model for an original dimension x .

Question 8

Order and context

No calculator

Compare $B(S(x))$ with $S(B(x))$ from the previous question and interpret the difference.

Question 9

Error analysis

No calculator

A student writes $f(g(x)) = f(x)g(x)$. Give a counterexample using $f(x) = x + 1$ and $g(x) = 2x$.

Question 10

Domain

No calculator

Find the domain of $\ln(4 - x^2)$.

Question 11

Graph structure

No calculator

Let $f(x) = 2^x$ and $g(x) = x - 3$. Describe how $f \circ g$ transforms the graph of f , and compare with $g \circ f$.

Question 12

Construction

No calculator

Construct functions f and g so that $(f \circ g)(x) = \frac{1}{(x+2)^2 + 5}$, with g polynomial and f rational.

Section C: AP-style multiple choice**MCQ 1**

No calculator

If $f(x) = x^2 - 1$ and $g(x) = x + 3$, what is $(f \circ g)(x)$?

- A. $x^2 + 2$
- B. $x^2 + 6x + 8$
- C. $x^2 + 3$
- D. $(x^2 - 1)(x + 3)$

MCQ 2

No calculator

Which restriction belongs to the domain of $\sqrt{5 - 2x}$?

- A. $x \geq 5/2$
- B. $x \leq 5/2$
- C. $x \neq 5/2$
- D. All real x

MCQ 3

No calculator

Which statement is always true?

- A. $f \circ g = g \circ f$
- B. $f(g(x)) = f(x)g(x)$
- C. The output of the inner function must be in the outer function's domain.
- D. Every function can be decomposed uniquely.

MCQ 4

No calculator

If $f(g(4)) = 9$ and $g(4) = 2$, which statement follows?

- A. $f(4) = 9$
- B. $f(2) = 9$
- C. $g(9) = 4$
- D. $g(2) = 9$

Section D: Free-response reasoning**FRQ 1 - Formula, domain, and order**

No calculator

6 points

Let $f(u) = \sqrt{u-2}$ and $g(x) = 6 - x^2$. (a) Find $(f \circ g)(x)$ and its domain. (b) Find $(g \circ f)(x)$ and its domain. (c) Evaluate both compositions at $x = 2$ when defined. (d) Explain why the domains and formulas differ.

FRQ 2 - Composite process in context

Calculator permitted

6 points

A laboratory converts a raw sensor reading x to concentration by $C(x) = 0.8x - 1.5$ mg/L. A reporting system then converts concentration c to a risk score by $R(c) = 12 \ln(c + 1)$. (a) Write the risk score as a function of the raw reading. (b) Determine its domain. (c) Find the score for $x = 5$. (d) Explain why reversing the order is not meaningful in this context.
